

FRACTIONS

Short Revision Notes | Analyze Level (Break Information into Parts and Identify Relationships)

1. Breaking Down a Fraction

Component	What It Represents	Example: 3/4
Numerator (3)	Number of parts taken	3 parts are shaded/taken
Denominator (4)	Total number of equal parts	The whole is divided into 4 equal parts
Fraction Bar	Division symbol	3/4 means 3 / 4
Value	Part / Whole	3 / 4 = 0.75

Relationship Analysis:

Fraction = Numerator / Denominator
 $3/4 = 3 / 4 = 0.75$

If numerator increases -> fraction value increases
If denominator increases -> fraction value decreases
If numerator = denominator -> fraction = 1
If numerator > denominator -> fraction > 1 (improper)

2. Comparing Fractions - Relationship Analysis

Methods to Compare:

Method	How To	Example (Compare $\frac{2}{3}$ and $\frac{3}{4}$)
Same Denominator	Convert to common denominator	$\frac{2}{3} = \frac{8}{12}$, $\frac{3}{4} = \frac{9}{12} \rightarrow \frac{3}{4} > \frac{2}{3}$
Cross Multiplication	$\frac{a}{b} ? \frac{c}{d} \rightarrow a \times d ? b \times c$	$\frac{2}{3} ? \frac{3}{4} \rightarrow 2 \times 4 ? 3 \times 3 \rightarrow 8 ? 9 \rightarrow \frac{3}{4} > \frac{2}{3}$
Decimal	Convert to decimals	$\frac{2}{3} = 0.666$, $\frac{3}{4} = 0.75 \rightarrow \frac{3}{4} > \frac{2}{3}$

Comparison Analysis Table:

Fraction 1	Fraction 2	Cross Multiply	Which is Larger?
$\frac{2}{3}$	$\frac{3}{4}$	$2 \times 4 = 8$, $3 \times 3 = 9$	$\frac{3}{4} > \frac{2}{3}$
$\frac{3}{5}$	$\frac{4}{5}$	$3 \times 5 = 15$, $5 \times 4 = 20$	$\frac{4}{5} > \frac{3}{5}$
$\frac{5}{6}$	$\frac{7}{8}$	$5 \times 8 = 40$, $6 \times 7 = 42$	$\frac{7}{8} > \frac{5}{6}$
$\frac{7}{10}$	$\frac{2}{3}$	$7 \times 3 = 21$, $10 \times 2 = 20$	$\frac{7}{10} > \frac{2}{3}$

Key Relationship

- The larger the numerator relative to denominator, the larger the fraction.

3. Analyzing Equivalent Fractions

Relationship Pattern:

Original	x2	x3	x4	Pattern
1/2	2/4	3/6	4/8	Numerator and denominator grow proportionally
1/3	2/6	3/9	4/12	Both multiply by same number
3/4	6/8	9/12	12/16	Ratio remains constant (3:4)

Simplification Analysis:

Fraction	Divide by HCF	Simplified	Relationship
4/6	2	2/3	Both divide by 2
6/9	3	2/3	Both divide by 3
8/12	4	2/3	Both divide by 4
12/16	4	3/4	Both divide by 4
10/25	5	2/5	Both divide by 5

Key Analysis

- Equivalent fractions have a CONSTANT RATIO between numerator and denominator.

4. Analyzing Improper Fractions and Mixed Numbers

Relationship Between Improper and Mixed:

Improper	Division	Mixed	Parts Analysis
$7/3$	$7 \div 3 = 2 \text{ R}1$	$2 \frac{1}{3}$	2 wholes + $1/3$
$11/4$	$11 \div 4 = 2 \text{ R}3$	$2 \frac{3}{4}$	2 wholes + $3/4$
$17/5$	$17 \div 5 = 3 \text{ R}2$	$3 \frac{2}{5}$	3 wholes + $2/5$
$27/8$	$27 \div 8 = 3 \text{ R}3$	$3 \frac{3}{8}$	3 wholes + $3/8$

Number Line Analysis:

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0 1 2 3 4
|----|----|----|----|
^  ^  ^
5/3 7/3 11/3
(1 2/3)(2 1/3)(3 2/3)

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Observation: Improper fractions are between whole numbers

5. Analyzing Addition and Subtraction

With Same Denominator (Relationship Focus):

Expression	Parts Analysis	Result
$\frac{2}{7} + \frac{3}{7}$	2 parts + 3 parts = 5 parts of size $\frac{1}{7}$	$\frac{5}{7}$
$\frac{5}{6} - \frac{2}{3}$	5 parts - 2 parts = 3 parts of size $\frac{1}{6}$	$\frac{3}{6} = \frac{1}{2}$

With Different Denominators:

Expression	Step 1: Find LCM	Step 2: Convert	Step 3: Add/Subtract
$\frac{1}{2} + \frac{1}{3}$	$\text{LCM}(2,3)=6$	$\frac{3}{6} + \frac{2}{6}$	$\frac{5}{6}$
$\frac{3}{4} - \frac{2}{5}$	$\text{LCM}(4,5)=20$	$\frac{15}{20} - \frac{8}{20}$	$\frac{7}{20}$
$\frac{2}{3} + \frac{1}{4}$	$\text{LCM}(3,4)=12$	$\frac{8}{12} + \frac{3}{12}$	$\frac{11}{12}$
$\frac{5}{6} - \frac{1}{3}$	$\text{LCM}(6,3)=6$	$\frac{5}{6} - \frac{2}{6}$	$\frac{3}{6} = \frac{1}{2}$

Pattern Analysis

- Same denominator -> just add/subtract numerators
- Different denominator -> find COMMON ground (LCM) -> then add/subtract

6. Analyzing Multiplication of Fractions

Pattern Analysis:

Expression	Numerators	Denominators	Result
$\frac{1}{2} \times \frac{1}{3}$	$1 \times 1 = 1$	$2 \times 3 = 6$	$\frac{1}{6}$
$\frac{2}{3} \times \frac{3}{4}$	$2 \times 3 = 6$	$3 \times 4 = 12$	$\frac{6}{12} = \frac{1}{2}$
$\frac{2}{5} \times \frac{3}{5}$	$2 \times 3 = 6$	$5 \times 5 = 25$	$\frac{6}{25}$
$\frac{3}{4} \times \frac{4}{5}$	$3 \times 4 = 12$	$4 \times 5 = 20$	$\frac{12}{20} = \frac{3}{5}$

Key Observation

- Multiplying proper fractions -> RESULT IS SMALLER than each factor
- $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ ($\frac{1}{6} < \frac{1}{2}$ and $\frac{1}{6} < \frac{1}{3}$)
- $\frac{3}{4} \times \frac{4}{5} = \frac{3}{5}$ ($\frac{3}{5} < \frac{3}{4}$ and $\frac{3}{5} < \frac{4}{5}$)

Why? You're taking a "part of a part" — $\frac{1}{2} \times \frac{1}{3} =$ half of a third — even smaller.

7. Analyzing Division of Fractions

Pattern Analysis:

Expression	Reciprocal	Multiplication	Result
$1/2 \div 1/3$	$3/1$	$1/2 \times 3$	$1 \frac{1}{2}$
$3/4 \div 1/2$	$2/1$	$3/4 \times 2$	$1 \frac{1}{2}$
$2/3 \div 3/4$	$4/3$	$2/3 \times 4/3$	$8/9$
$1/5 \div 2/5$	$5/2$	$1/5 \times 5/2$	$1/2$

Key Observation

- Dividing by a proper fraction -> RESULT IS LARGER than the first fraction
- $1/2 \div 1/3 = 1 \frac{1}{2}$ ($1 \frac{1}{2} > 1/2$)
- $3/4 \div 1/2 = 1 \frac{1}{2}$ ($1 \frac{1}{2} > 3/4$)

Why? Division asks "How many times does the divisor fit into the dividend?" $1/2 \div 1/3 =$ "How many thirds fit in half?" -> $1 \frac{1}{2}$ times

8. Relationship Between Fractions, Decimals, and Percentages

Fraction	Division	Decimal	x100	Percentage
1/2	1/2	0.5	0.5x100	50%
1/4	1/4	0.25	0.25x100	25%
3/4	3/4	0.75	0.75x100	75%
1/5	1/5	0.2	0.2x100	20%
3/5	3/5	0.6	0.6x100	60%
4/5	4/5	0.8	0.8x100	80%

Pattern Analysis:

Fraction -> Divide -> Decimal -> Multiply by 100 -> Percentage
5/8 -> $5/8 = 0.625$ -> $0.625 \times 100 = 62.5\%$

9. Analyzing the "OF" Relationship

Expression	Interpretation	Operation	Result
1/2 of 100	Half of 100	$1/2 \times 100$	50
1/3 of 20	One third of 20	$1/3 \times 20$	$6 \frac{2}{3}$
1/4 of 3/4	Quarter of three quarters	$1/4 \times 3/4$	$3/16$
2/3 of $1 \frac{1}{2}$	Two thirds of one and a half	$2/3 \times 3/2$	1

Pattern Analysis

- "of" ALWAYS means multiplication
- $1/2$ of 100 = $1/2 \times 100 = 50$
- $1/3$ of $1/5 = 1/3 \times 1/5 = 1/15$

10. Analyzing Reciprocals

Relationship Pattern:

Number	Reciprocal	Product	Relationship
$2/3$	$3/2$	1	They are inverses
$3/4$	$4/3$	1	They are inverses
5	$1/5$	1	They are inverses
$1 \frac{1}{2} = 3/2$	$2/3$	1	They are inverses

Key Understanding

- Reciprocal is the “mirror image” of a fraction
- When you multiply a number by its reciprocal, you get 1
- This is why division becomes multiplication by the reciprocal

11. Analyzing the BODMAS Relationship

Order of Operations Analysis:

Priority	Operation	Example	Why?
1st	Brackets ()	$2(3+1) = 8$	Most important
2nd	Of	$1/2 \text{ of } 10 = 5$	Before / and x
3rd	Division	$6 / 2 = 3$	Left to right
3rd	Multiplication	$3 \times 4 = 12$	Left to right
4th	Addition	$5 + 3 = 8$	Left to right
4th	Subtraction	$8 - 2 = 6$	Left to right

Analyzing the Relationship:

Expression	Without BODMAS	With BODMAS	Correct Answer
$2 + 3 \times 4$	$5 \times 4 = 20$	$2 + 12 = 14$	14
$1/2 \times 3 + 1$	$1/2 \times 4 = 2$	$1 \frac{1}{2} + 1 = 2 \frac{1}{2}$	$2 \frac{1}{2}$
$6 / 2 \times 3$	$6 / 6 = 1$	$3 \times 3 = 9$	9

Observation

- BODMAS ensures everyone gets the SAME answer!

12. Comparing Different Types of Fractions

Type	Numerator	Denominator	Value	Example
Proper	Smaller	Larger	< 1	$2/3$, $3/4$, $4/5$
Improper	Larger or Equal	Smaller or Equal	≥ 1	$5/3$, $7/4$, $3/3$
Unit	1	Any	$1/n$	$1/2$, $1/3$, $1/4$
Like	Any	Same	Compare easily	$2/7$, $3/7$
Unlike	Any	Different	Need LCM	$2/3$, $4/5$

13. Identifying Relationships in Word Problems

Problem Type	Key Words	Operation	Fraction Relationship
Finding a part	"of", "fraction of"	Multiplication	Part = Fraction $\times$ Whole
Finding remaining	"left", "remaining"	Subtraction	Whole - Used = Remaining
Finding total	"together", "total"	Addition	Part + Part = Whole
Sharing equally	"divided into", "each"	Division	Total / Number = Share

Word Problem Analysis:

Problem	Identify	Relationship	Answer
Spent $\frac{1}{4}$ on food, $\frac{1}{2}$ on business	$\frac{1}{4} + \frac{1}{2}$ used	Remaining = $1 - (\frac{1}{4} + \frac{1}{2})$	$\frac{1}{4}$ saved
Gave $\frac{1}{2}$ to son, $\frac{1}{3}$ to daughter	$\frac{1}{2} + \frac{1}{3}$ given away	Total given = $\frac{1}{2} + \frac{1}{3}$	$\frac{5}{6}$ given away
Son donates $\frac{1}{5}$ of his portion	$\frac{1}{5}$ of $\frac{1}{2}$	Donation = $\frac{1}{5} \times \frac{1}{2}$	$\frac{1}{10}$ of total

14. Analyzing Simplification

Simplifying Process:

Fraction	Common Factor	Division	Simplified Form
4/6	2	4/2, 6/2	2/3
6/9	3	6/3, 9/3	2/3
12/16	4	12/4, 16/4	3/4
20/25	5	20/5, 25/5	4/5
21/28	7	21/7, 28/7	3/4

Relationship Analysis

- Both numerator AND denominator must be divided by the SAME number
- The value of the fraction does NOT change
- Simplified form has the SMALLEST possible numerator and denominator

15. Analyzing Common Mistakes

Mistake	Analysis	Why It's Wrong
$1/2 + 1/3 = 2/5$	Adding numerators and denominators separately	You can only add when denominators are equal
$1/2 \times 1/2 = 1/4$	Actually this is CORRECT!	-
$1/2 \div 1/3 = 1/6$	Dividing by a fraction gives LARGER result	Should be $1 \frac{1}{2}$
$3/3 = 0$	$3/3$ equals 1	Three thirds make a whole
Improper fractions are "wrong"	Improper fractions are acceptable	They represent values ≥ 1

16. Analysis Checklist

When analyzing a fraction:

- What is the numerator? What does it represent?
- What is the denominator? What does it represent?
- Is it proper, improper, or a unit fraction?
- What is its decimal equivalent?
- What is its percentage equivalent?

When comparing fractions:

- Do they have the same denominator?
- Use cross multiplication: $a \times d$? $b \times c$
- Convert to decimals: which is larger?

When adding/subtracting:

- Are denominators the same?
- If not, what is the LCM?
- Convert to equivalent fractions
- Add/subtract numerators

When multiplying:

- What patterns do you notice?
- Is the result larger or smaller than each factor?
- Can you cancel before multiplying?

When dividing:

- What is the reciprocal of the divisor?
- Why is the result larger than the first fraction?

When using BODMAS:

- What operations are involved?
- What is the correct order?
- Are there any "of" operations?

When simplifying:

- What is the HCF of numerator and denominator?
- Can it be divided further?
- Is it in simplest form?

17. Self-Test - Analysis Questions

No.	Question	Answer
1	In $\frac{5}{8}$, what does 5 represent?	Number of parts taken
2	In $\frac{5}{8}$, what does 8 represent?	Total number of equal parts
3	Which is larger: $\frac{2}{3}$ or $\frac{3}{4}$?	$\frac{3}{4}$ (cross multiply: $8 < 9$)
4	Why is $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$?	You're taking half of one third
5	Why is $\frac{1}{2} \div \frac{1}{3} = 1 \frac{1}{2}$?	One and a half thirds fit in half
6	Why are $\frac{1}{2}$ and $\frac{2}{4}$ equivalent?	Both represent the same portion
7	What happens to a fraction when numerator increases?	The value increases
8	What happens to a fraction when denominator increases?	The value decreases
9	Why is BODMAS important?	Ensures everyone gets the same answer
10	What is "of" equivalent to?	Multiplication (x)

Revision Tip: Analyzing fractions means understanding HOW and WHY they work. Don't just memorize rules — ask yourself “Why does this work?” This deeper understanding will help you solve any fraction problem!